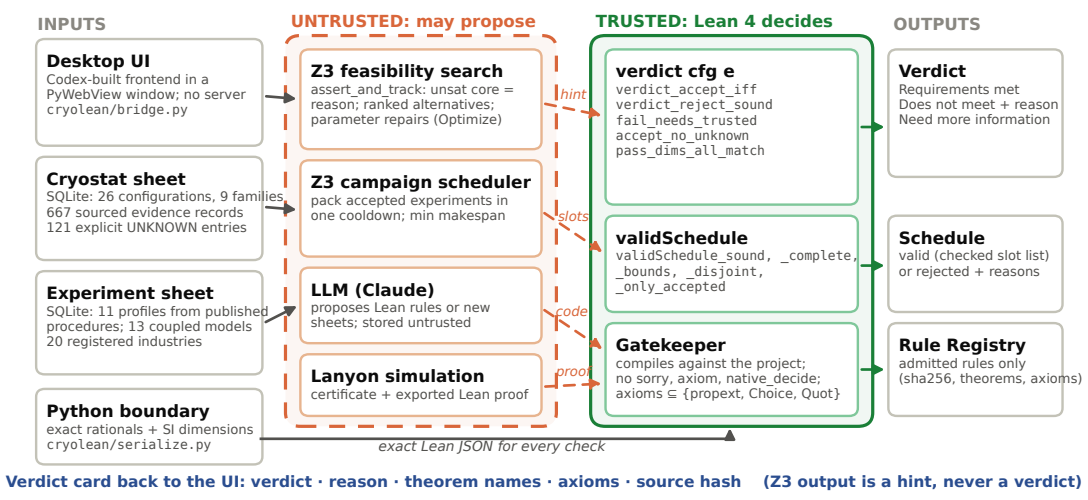
INPUTS
Desktop UI
Codex-built frontend in a PyWebView window; no server
cryolean/bridge.py
Cryostat sheet
SQLite: 26 configurations, 9 families
667 sourced evidence records
121 explicit UNKNOWN entries
Experiment sheet
SQLite: 11 profiles from published procedures; 13 coupled models
20 registered industries
Python boundary
exact rationals + SI dimensions
cryolean/serialize.py
UNTRUSTED: may propose
Z3 feasibility search
assert_and_track: unsat core = reason; ranked alternatives; parameter repairs (Optimize)
Z3 campaign scheduler
pack accepted experiments in one cooldown; min makespan
LLM (Claude)
proposes Lean rules or new sheets; stored untrusted
Lanyon simulation
certificate + exported Lean proof
hint
slots
code
proof
TRUSTED: Lean 4 decides
verdict cfg e
verdict_accept_iff
verdict_reject_sound
fail_needs_trusted
accept_no_unknown
pass_dims_all_match
validSchedule
validSchedule_sound, _complete, _bounds, _disjoint, _only_accepted
Gatekeeper
compiles against the project; no sorry, axiom, native_decide; axioms ⊆ {propext, Choice, Quot}
OUTPUTS
Verdict
Requirements met
Does not meet + reason
Need more information
Schedule
valid (checked slot list) or rejected + reasons
Rule Registry
admitted rules only (sha256, theorems, axioms)
exact Lean JSON for every check
Verdict card back to the UI: verdict · reason · theorem names · axioms · source hash (Z3 output is a hint, never a verdict)